

DAY 00 / MASTER INDEX

Foundation Collection Learning Index

A guided map of 28 beginner-friendly Python lessons built around practical password, banking, profile, and transaction exercises.

28

DAILY LESSONS

215

PDF PAGES

4

LEARNING WEEKS

RECOMMENDED PATH

1**Learn values and decisions**

Days 1-7: variables, input/output, conversion, conditions, validation, and revision.

2**Build repeatable programs**

Days 8-14: menus, while and for loops, lists, functions, and return values.

3**Model and clean data**

Days 15-21: string methods, slicing, dictionaries, nested records, and sets.

4**Combine practical patterns**

Days 22-28: counters, search, aggregation, tuples, dictionary iteration, and error handling.

WEEKLY RHYTHM

Complete six focused lessons, then use every seventh day to revise, connect the concepts, and solve a combined exercise.

WEEK 01 / DAYS 01-07

Python basics and banking decisions

Start with variables, strings, input and output, then use conditions and validation to model simple banking operations.

DAY 01	Save Password FOCUS variables; comments; indentation; input(); output PDF Day-01-Save-Password.pdf	5 PAGES
DAY 02	Validate Password FOCUS strings; ==; Boolean thinking; if-else PDF Day-02-Validate-Password.pdf	6 PAGES
DAY 03	Change Password FOCUS password comparison; nested decisions; empty-input validation PDF Day-03-Change-Password.pdf	6 PAGES
DAY 04	Open Bank Account FOCUS strings; numbers; type conversion; functions PDF Day-04-Open-Bank-Account.pdf	6 PAGES
DAY 05	Deposit Amount FOCUS addition; +=; balance updates; input validation PDF Day-05-Deposit-Amount.pdf	8 PAGES
DAY 06	Withdraw Amount with a Limit FOCUS if-elif-else; comparisons; multiple conditions; validation PDF Day-06-Withdraw-Amount-with-Limit.pdf	8 PAGES
DAY 07	Weekly Revision: Days 1-6 FOCUS variables; input/output; conversion; conditions; operators PDF Day-07-Weekly-Revision.pdf	12 PAGES
MEMORY LINE Input enters as text. Convert only when needed, validate before changing state, and make each decision branch explicit.		

WEEK 02 / DAYS 08-14

Menus, loops, lists, and functions

Move from one-time decisions to repeated programs, collections of values, numbered menus, and reusable functions.

DAY 08	Simple Menu Choice FOCUS menus; input(); integer conversion; if-elif-else PDF Day-08-Simple-Menu-Choice.pdf	7 PAGES
DAY 09	Repeat a Menu Using a while Loop FOCUS while; break; loop conditions; repeating menus PDF Day-09-Repeat-Menu-Using-While-Loop.pdf	9 PAGES
DAY 10	Print Numbered Options Using range() FOCUS for; range(); lists; indexing PDF Day-10-Print-Numbered-Options-Using-range.pdf	7 PAGES
DAY 11	Store a Transaction in a List FOCUS lists; append(); string formatting PDF Day-11-Store-Transaction-in-List.pdf	7 PAGES
DAY 12	Show Transactions Using a Loop FOCUS lists; for; empty-list checks; functions PDF Day-12-Show-Transactions-Using-Loop.pdf	6 PAGES
DAY 13	Create a Deposit Function FOCUS def; parameters; return; function calls PDF Day-13-Deposit-Function.pdf	7 PAGES
DAY 14	Weekly Revision: Days 8-13 FOCUS menus; loops; range(); lists; functions; return PDF Day-14-Weekly-Revision.pdf	13 PAGES

MEMORY LINE

Use loops for repetition, lists for ordered data, and functions for a single reusable responsibility with clear inputs and outputs.

WEEK 03 / DAYS 15-21

Strings and data collections

Clean and transform text, then choose dictionaries, nested records, or sets according to the structure of the data.

DAY 15	Clean an Account Holder Name FOCUS strip(); title(); lower(); upper() PDF Day-15-Clean-Account-Holder-Name.pdf	7 PAGES
DAY 16	Mask an Account Number Using Slicing FOCUS indexing; slicing; negative indexes; len(); repetition PDF Day-16-Mask-Account-Number-Using-Slicing.pdf	9 PAGES
DAY 17	Clean a Phone Number FOCUS replace(); strip(); method chaining; string validation PDF Day-17-Clean-Phone-Number.pdf	8 PAGES
DAY 18	Create and Update a Profile Dictionary FOCUS key-value pairs; access; update; .get() PDF Day-18-Create-and-Update-Profile-Dictionary.pdf	8 PAGES
DAY 19	Add a Payee Using a List of Dictionaries FOCUS lists; dictionaries; append(); records PDF Day-19-Add-Payee-Using-List-of-Dictionaries.pdf	7 PAGES
DAY 20	Store Unique Cities Using a Set FOCUS sets; uniqueness; duplicate removal; add(); conversion PDF Day-20-Unique-Cities-Using-Set.pdf	6 PAGES
DAY 21	Weekly Revision: Days 15-20 FOCUS string methods; slicing; dictionaries; nested collections; sets PDF Day-21-Weekly-Revision.pdf	13 PAGES

MEMORY LINE

Strings are transformed into new values. Dictionaries model named fields, lists hold ordered records, and sets enforce uniqueness.

WEEK 04 / DAYS 22-28

Search, aggregation, records, and errors

Combine the earlier building blocks into practical patterns for limited attempts, searching, totals, immutable records, iteration, and safe input.

DAY 22	Limited Login Attempts FOCUS while; counters; loop conditions; break; comparisons PDF Day-22-Limited-Login-Attempts.pdf	6 PAGES
DAY 23	Search for a Payee by Name FOCUS loops; dictionary access; Boolean flags; search patterns PDF Day-23-Search-Payee-by-Name.pdf	8 PAGES
DAY 24	Calculate Total Deposits FOCUS lists; for; accumulators; running totals; return PDF Day-24-Calculate-Total-Deposits.pdf	6 PAGES
DAY 25	Create a Transaction Tuple Record FOCUS tuples; indexing; fixed data PDF Day-25-Transaction-Tuple-Record.pdf	6 PAGES
DAY 26	Print a Dictionary Using items() FOCUS .items(); keys; values; for loops PDF Day-26-Print-Dictionary-Using-items.pdf	6 PAGES
DAY 27	Handle Amount Input Safely FOCUS try; except; float(); ValueError; friendly errors PDF Day-27-Safe-Amount-Input.pdf	6 PAGES
DAY 28	Weekly Revision: Days 22-27 FOCUS counters; searching; accumulators; tuples; .items(); try-except PDF Day-28-Weekly-Revision.pdf	12 PAGES
MEMORY LINE Track loop state deliberately, use an accumulator for totals, select immutable tuples for fixed records, and catch only errors you can handle.		

QUICK LOCATOR

Find the right lesson fast

Use this map when you know the Python concept you want to practice but not the day that covers it.

Want to practice...	Core concepts	Go to
Input and output	input(), print(), variables, conversion	Days 1, 4
Conditions and validation	if-else, elif, comparisons, nested decisions	Days 2-6, 8
Loops and counters	while, for, range(), break, loop state	Days 9-10, 12, 22
Lists and accumulation	append(), traversal, totals, ordered records	Days 10-12, 19, 24
Reusable functions	def, parameters, return, clear responsibilities	Days 4, 12-13, 24
String cleaning	strip(), case methods, replace(), slicing	Days 15-17
Dictionaries and records	key-value access, update, .get(), .items()	Days 18-19, 23, 26
Unique or fixed data	sets, duplicate removal, tuples	Days 20, 25
Safe numeric input	try-except, float(), ValueError	Day 27
Weekly consolidation	combined practice and recall	Days 7, 14, 21, 28

COMPLETION CHECK

Understand	Explain the concept and predict the output before running code.
Build	Write the exercise without copying, using meaningful names and clean indentation.
Test	Try normal, empty, boundary, duplicate, and invalid inputs where relevant.
Reflect	State one mistake to avoid and one place the pattern applies in a real program.